

Comprehensive Test Case Document

Test Cases: QueryString

Input	Action	Expected Result
New QueryString('name=John&age=25&country=UK')	getParameter('name')	Returns 'John'
New QueryString('name=John&age=25&country=UK')	getParameter('age')	Returns '25'
New QueryString('name=John&age=25&country=UK')	getParameter('country')	Returns 'UK'
New QueryString('name=John&age=25')	getParameter('email')	Returns null (parameter not found)
New QueryString('message=Hello+World&url=https%3A%2F%2Fexample.com')	getParameter('message')	Returns 'Hello World'
New QueryString('message=Hello+World&url=https%3A%2F%2Fexample.com')	getParameter('url')	Returns 'https://example.com'
New QueryString('name=John&age=25&empty=')	hasParameter('empty')	True (exists but value empty)
New QueryString('name=John&age=25')	hasParameter('email')	False (parameter does not exist)
New QueryString('name=John&age=25&last=Doe')	getParameterOffset('name')	0
New QueryString('name=John&age=25&last=Doe')	getParameterOffset('age')	10
New QueryString('name=John&age=25&last=Doe')	getParameterOffset('last')	17
QueryString.decode('Hello+World')	decode()	Returns 'Hello World'
QueryString.decode('100%25+happy')	decode()	Returns '100% happy'
QueryString.decode('%40%23%24')	decode()	Returns '@#\$'
QueryString.decode('++')	decode()	Returns two spaces ' '

Test Cases: VendingMachine

Input	Action	Expected Result
New VendingMachine()	getBalance()	Returns 0 (initial balance)
New VendingMachine()	getRevenue()	Returns 0 (initial revenue)
Insert 25, Insert 10	getBalance()	Returns 35 (sum of coins inserted)
Insert 25, Insert 25, then refund()	refund()	Returns 50 refunded and resets balance to 0
Insert 1	vendCandyBar()	Returns False (insufficient balance), balance=1, revenue=0
Insert 3	vendCandyBar()	Returns True (successful vend), balance=0, revenue=3
Perform 20 successful vends (3 coins each)	vendCandyBar()	Depletes stock (20 candy bars sold)
After stock empty	vendCandyBar()	Returns False (out of stock)
After restock()	vendCandyBar()	Returns True (machine restocked successfully)
After 21 total successful vends	getRevenue()	Returns 63 (21 x 3 coins each)